

Archive For / In the Future

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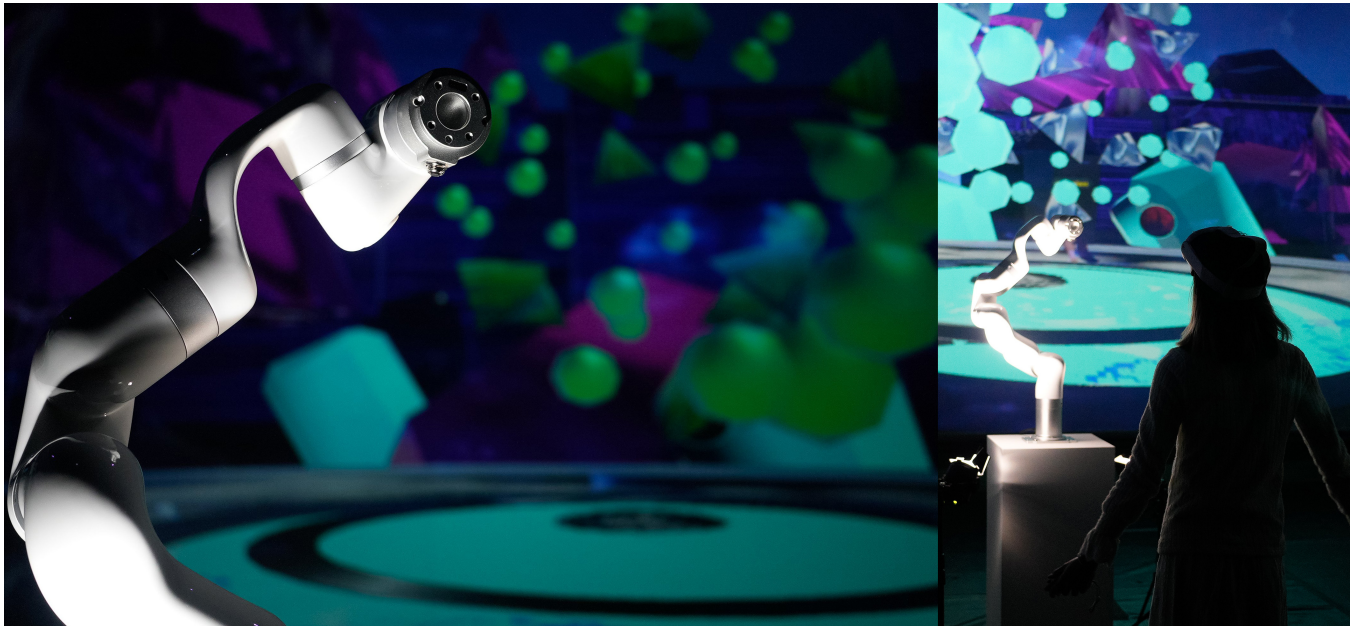


Figure 1: Archive For / In the Future is a VRChat environment that uses a physical robot to represent movement gestures for a potential future. Our memories of these actions have been transfigured through time, as the archive deviates more and more from historical past through reinterpretation. What remains is the form of the interactions captured in a digital reality.

Abstract

When we visit museums, we see reconstructions of ancient sites that speak more about why we interpret them this way rather than how they really looked. We build models about an ancient places that we have never been by interpreting history through our own biased lens, imagining their daily lives as if they lived with our own social customs. Our interpretations of the distant past are wrought with speculation and imagination, because every time we bring that record up from memory, we reinterpret them anew in the context of our own biases today. Is this also true for

how the future will interpret us? Will they look at us from the lens of a distant future that no longer understands us for who we are today? Will they even speak the same language as us, but rather diverge in motives and priorities so radically that they will fail to recognize us? To show the need for preserving memories and interpretations as part of a collective heritage for the future, we created a physical installation with robot arm that serves as an archive from the future that attempts to remember our present. We used ChatGPT to create personas from history and converted their conversational snippets into a non-human language of the future in the form of robotic movements. The robot has a virtual existence in VRChat, where visitors can engage with the physical archive presence online, showing how the future can choose to remember us through an imagined interactive world living online. This work explores how hybrid physical-digital presence can illustrate our precarious interpretations of the past and present, inspiring us to preserve intangible forms of our current heritage in consideration for continuously changing customs of the future.

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CCS Concepts

• Applied computing → Media arts.

Keywords

cultural heritage, robotic art, VRChat, ChatGPT, speculative design

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1 Project URLs

Website: <https://archivefuture.wordpress.com/>

Review: <https://archivefuture.wordpress.com/review-lee/>

Interview video: <https://youtu.be/aFJfMrf0XXM>

2 Description of Artwork

Our remembrances of the past are fought with interpretations, inaccuracies, and misperceptions[He et al. 2025]. How may the future interpret us in the present if we fail to represent the past? This work creates an archive in the form of immersive VRChat space and robotic movements that speak gesture language of the future[Chang et al. 2025].

ARCHIVE FOR / IN THE FUTURE explores how we preserve the past from the vantage point of the future, using GTP-based interpretations of a city's history to illustrate how our technologies may determine our precarious interpretations of the past. By translating GPT interpretations into movement language of the future driven by robotic arm[LC et al. 2023a, 2024], we invite visitors to renew our interpretations of this past, and to explore how our current technologies determine what we can remember about that past[Fu et al. 2024; Miller et al. 2025]. The robotic interactions are reminiscent of real human interactions, but they also give a sense of the uncanny, resulting from a civilization that we don't seem to understand and yet appear to be familiar with, a society that attempts to remember from our future. This work dares us to recall the past, and realize how our own technologies of the present remake the past for the future[Wu et al. 2024].

Our previous exploration of human interpretations in cultural heritage developed the idea of human memories and imaginations as part of an intangible form of heritage[LC 2024]. That work indicated how our personal memories may be subject to reinterpretation and change over time[Zhang et al. 2025a]. The present work takes the consideration into the future, asking how the interpretations would change when a future entity is tasked to remember our present. The technologies we develop in GenAI and robotics are meant to serve a narrative, showing how we express our deepest desires and fears even when interacting in new media[Zhou et al. 2025]. The media may change, but the universal human stories remain.

Our work explores how we continuously update at the present for consideration of a changing future, both in forms of our collective memory, and in the hopes and desires we imbue into our technologies of the present.

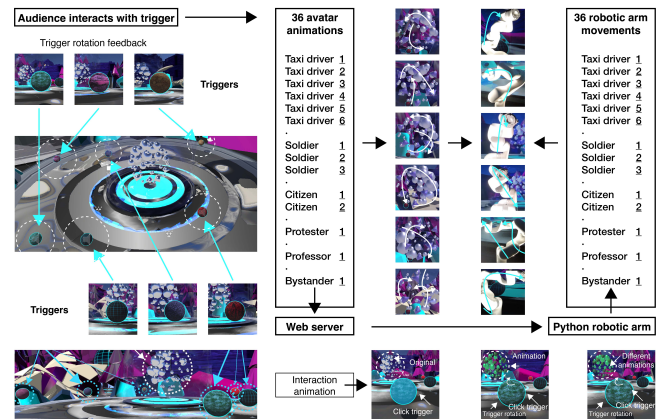


Figure 2: Using GPT to create characters from a historical perspective, then converting the generated narrative conversations into robot arm motions that have an analogous motion in an virtual space avatar.

3 Technical Information

GPT was used to represent characters from historical Gwangju by iterative prompting for specific style and narrative argument[Zeng et al. 2025]. We asked GPT to create conversations between the 6 characters specified², and mapped the selected textual snippets into robot motions[Dong et al. 2024] by prototyping with text-to-image tools[Han et al. 2024]. The motion-captured arm actions[Zhang et al. 2025b] are translated into avatar motions of embodied spheres[Chang et al. 2024; LC et al. 2023b] and deployed into a custom VRChat environment modeled on speculative designed visions [Gong et al. 2025] of Gwangju.

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